

TACHENG, China

WMO: 511330

Lat: 46.7320N

Lon: 82.9798E

Elev: 1755

StdP: 13.79

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.3	-5.4	-17.5	2.2	-8.6	-13.1	2.9	-3.9	18.8	22.1	15.9	22.4	3.2	20	0.487

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.2	94.0	65.2	91.2	64.3	88.4	63.5	68.2	86.4	66.7	84.5	65.4	82.8	5.4	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.5	90.2	73.1	60.8	85.0	72.6	59.1	80.0	72.7	33.6	86.4	32.3	84.5	31.3	82.7	75.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.8	14.0	11.4	DB	-15.8	99.7	5.7	2.9	-19.9	101.8	-23.2	103.4	-26.4	105.1	-30.5	107.2	(4)
(5)			WB	-16.2	70.6	5.6	2.5	-20.2	72.4	-23.5	73.8	-26.7	75.2	-30.7	77.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.3	14.3	19.9	34.8	52.0	62.5	71.7	74.6	72.5	62.4	48.7	33.0	19.5		
	DBStd	22.95	9.66	9.46	11.23	9.15	7.50	5.99	5.22	6.00	8.07	8.71	10.68	9.40		
	HDD50	4120	1108	842	483	84	5	0	0	0	8	129	514	947		
	HDD65	7347	1573	1262	938	394	137	14	2	10	139	506	960	1412		
	CDD50	3132	0	0	10	145	391	651	763	697	381	90	4	0		
	CDD65	883	0	0	0	5	58	215	299	242	62	2	0	0		
	CDH74	9634	0	0	0	85	684	2214	3201	2706	717	27	0	0		
	CDH80	3949	0	0	0	12	204	893	1437	1181	221	1	0	0		
(14) Wind	WSAvg	4.9	4.0	4.3	5.3	6.4	5.9	5.1	4.8	4.7	4.8	4.7	4.4	3.9		
(15) Precipitation	PrecAvg	11.1	0.6	0.5	0.7	1.2	1.1	1.1	1.2	0.6	0.5	1.3	1.3	0.8		
	PrecMax	18.4	1.7	1.3	2.5	3.5	3.1	4.8	3.6	2.2	1.9	3.0	3.7	2.0		
	PrecMin	6.1	0.0	0.1	0.1	0.0	0.2	0.4	0.1	0.0	0.0	0.0	0.3	0.1		
	PrecStd	3.0	0.4	0.3	0.6	0.9	0.8	0.8	1.0	0.6	0.4	0.8	0.9	0.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.1	44.4	67.9	81.9	90.6	94.8	98.6	97.6	91.1	77.7	62.7	41.2		
		MCWB	32.1	35.6	49.7	55.2	61.3	64.9	67.2	65.7	63.1	54.1	47.5	33.9		
	2%	DB	33.4	39.4	61.1	76.1	85.3	91.6	94.4	93.4	85.6	72.0	56.3	36.7		
		MCWB	29.9	32.5	46.3	53.4	58.8	64.3	66.2	64.9	59.7	51.7	44.2	31.4		
	5%	DB	30.6	35.9	56.0	72.0	81.4	88.7	91.1	90.2	82.0	67.4	50.7	34.0		
		MCWB	27.0	30.5	44.3	51.4	57.7	63.4	65.0	63.9	58.2	50.0	41.4	30.2		
	10%	DB	27.4	32.9	50.6	67.8	77.4	85.6	87.9	86.9	77.8	62.8	46.2	31.3		
		MCWB	24.1	28.6	41.6	50.1	56.4	62.6	64.5	62.8	56.3	48.3	39.6	28.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.9	36.5	51.2	58.2	63.7	68.7	71.5	69.3	64.1	56.0	49.6	35.4		
		MCDB	37.0	42.3	63.9	75.0	83.8	86.8	92.1	89.1	85.6	70.9	58.5	39.0		
	2%	WB	30.7	33.5	47.8	55.5	61.6	66.9	68.9	67.2	61.5	53.3	45.5	32.7		
		MCDB	32.9	37.9	58.9	69.8	78.1	85.2	87.1	86.3	82.0	67.9	53.1	35.5		
	5%	WB	27.1	31.2	45.1	53.5	59.9	65.5	67.2	65.5	59.5	51.5	42.6	30.6		
		MCDB	30.0	35.3	54.3	66.9	75.0	83.0	84.5	84.6	78.1	64.5	49.5	33.5		
	10%	WB	24.3	28.8	42.1	51.7	58.3	64.0	66.0	64.0	57.5	49.4	39.8	28.3		
		MCDB	27.3	32.7	50.5	64.0	72.8	80.6	82.6	82.6	74.6	60.8	46.0	31.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.4	17.1	17.0	20.1	21.3	21.8	22.2	23.8	23.5	20.2	15.1	14.6		
		MCDBR	17.5	19.5	23.0	26.3	26.4	25.9	27.0	28.0	28.0	26.9	21.8	15.8		
		MCWBR	13.6	14.2	12.5	11.5	10.1	9.4	9.6	10.8	12.1	12.8	12.3	11.6		
		MCDBR	15.4	17.3	21.2	21.1	21.7	22.5	22.8	24.2	25.5	23.6	19.7	13.6		
	5% WB	MCWBR	12.4	13.0	11.9	10.2	9.4	8.8	9.0	10.0	11.9	11.9	11.9	10.6		
(40) Clear-Sky Solar Irradiance	taub		0.307	0.338	0.385	0.425	0.406	0.411	0.414	0.385	0.358	0.356	0.339	0.307		
	taud		2.089	2.060	2.058	2.065	2.183	2.236	2.250	2.308	2.335	2.239	2.160	2.100		
	Ebn at Noon		237	255	261	262	272	271	268	272	270	248	223	220		
	Edh at Noon		30	38	44	49	45	43	42	38	34	31	28	26		
(44) All-Sky Solar Radiation	RadAvg		591	925	1302	1658	2066	2178	2090	1915	1507	934	527	436		
	RadStd		36	35	82	108	105	110	130	107	90	76	49	35		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page